MAGNETOM

Replacement of Parts System

Power distribution

Document Version

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Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad (TDK equipment type 8005 from 3M Deutschland GmbH).

WARNING

Please follow the General Safety Information (→ General Safety Notes / TD00-000.860.01) as well as the MR-Specific Safety Information (→ MR-specific safety information / MR-000.860.01)!

Error to comply with the Safety Information may result in death or serious injury.

» **Read the safety notes contained in the General Safety Information as well as the MR-Specific Safety Information in detail prior to starting work. Adhere to the information provided at all times.**

WARNING

Voltage continues to be present at the line voltage transformer even after the MR system has been switched off.

Failure to observe safety precautions may result in death or serious bodily injury.

» **Circuit breaker F100 in the line filter box (ACC, top) has to be switched off.**

WARNING

Voltage continues to be present at the line filter box even after the MR system and F100 have been switched off.

Failure to observe safety precautions may result in death or serious bodily injury.

» **Always set the on-site circuit breakers to OFF before performing any service activities on the line filter box.**

Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance <= 100 mOhm with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" and the protective earth wire connected to the "main component".

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

This chapter describes replacement procedures for the FRUs of the Power Distribution.

ACC				
FRU	**Time**	**Instructions**	**Final checks**	**Tools**
Circuit breaker (line filter box) F100, F102	45 min.	n.a.	n.a.	Normal tool set
Circuit breaker (ACC)	30 min.	n.a.	n.a.	Normal tool set
Contactor K1, K5, K2 K3, K4	45 min.	n.a.	n.a.	Normal tool set Allen key 5 mm
K10, K11, K13, K15, K17	45 min.	(→ Replacement of relay K10, K11, K13, K15, K17 / Page 8)	n.a.	Normal tool set Allen key, 5 mm
Power supply Vega 450	60 min	(→ Power supply (Lambda Vega 450) / Page 10)	n.a.	Normal tool set

3.1 Circuit breakers & contactors

3.1.1 Tools and time required

3.1.1.1 Time

20 min.

3.1.1.2 Tools

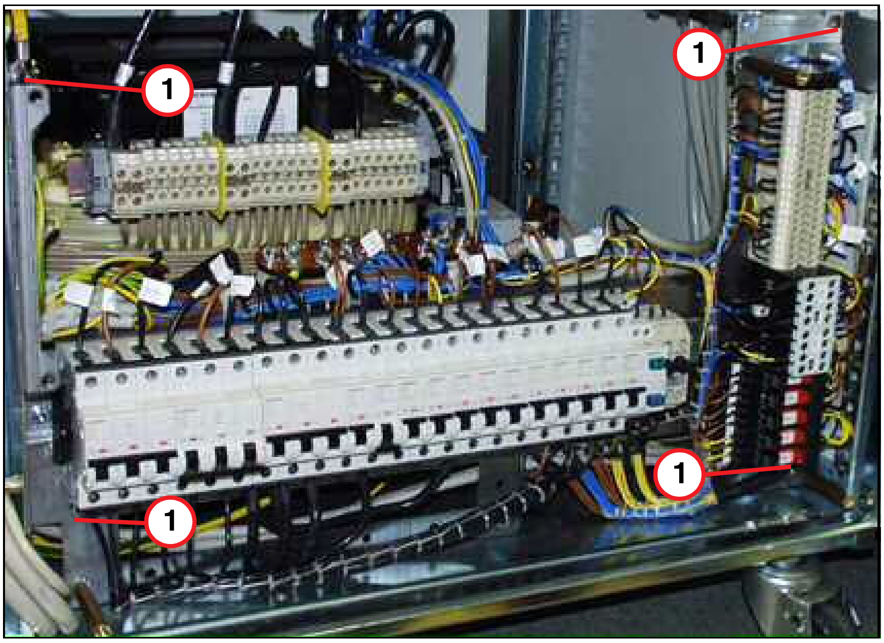
Normal tool set

3.1.2 Preliminary

The line voltage assembly can be tilted outward to improve access to the components.

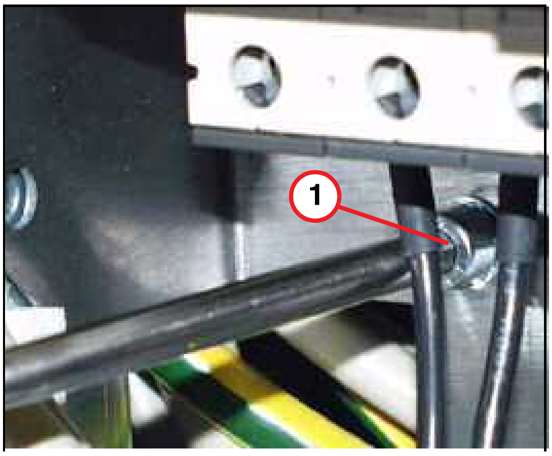
- Remove the cover
- Remove the screws for the rack

Fig. 1: Power voltage assembly



(1) Screw

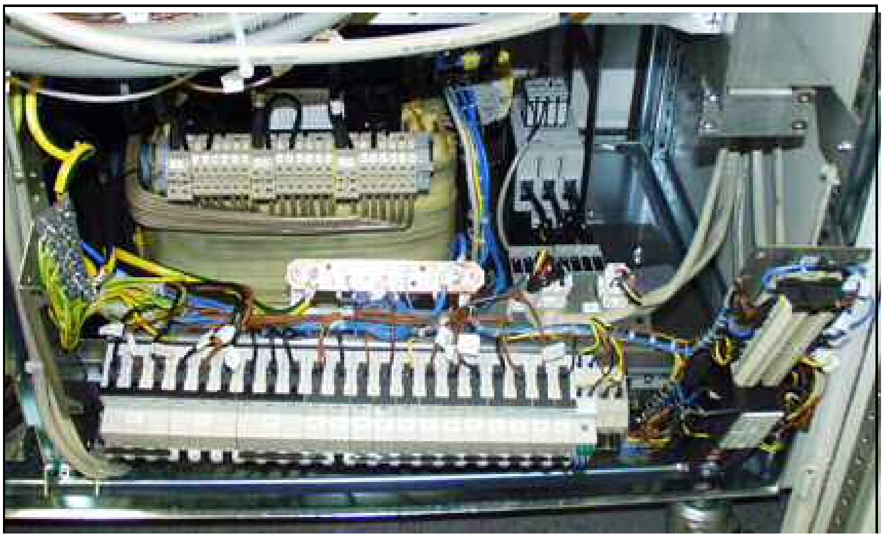
Fig. 2: Rack and screw



(1) Screw

- After removing the screws, the line voltage assembly can be tilted outward.

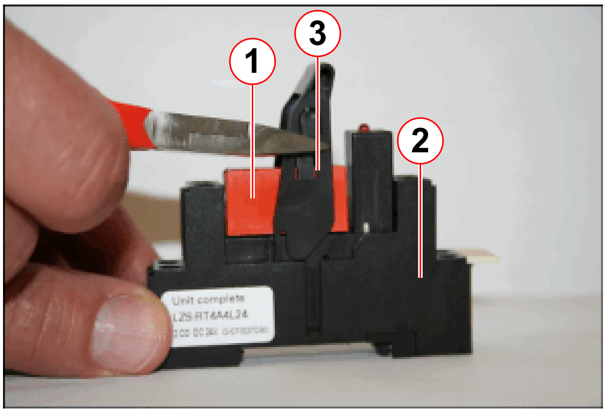
Fig. 3: Power voltage assembly



3.1.2.1 Replacement of relay K10, K11, K13, K15, K17

The relay numbers K10, K11, K13, K15, K17 have a new type of relay (LZS:RT4A4L24). The new relay is electrically compatible with the old relay type (LZX:RT4A4L24). The new relay can be extracted from the base (→ 1/Fig. 4 Page 9). The lever of the new relay is approx. 1 cm longer. Cut the lever of the base approx. 1 cm if the complete old relay needs to be replaced. (→ 3/Fig. 4 Page 9).

Fig. 4: Cutting the lever of the new relay type



(1) Relay
(2) Base
(3) Lever

3.2 Power supply (Lambda Vega 450)

3.2.1 Tools and time required

3.2.1.1 Time

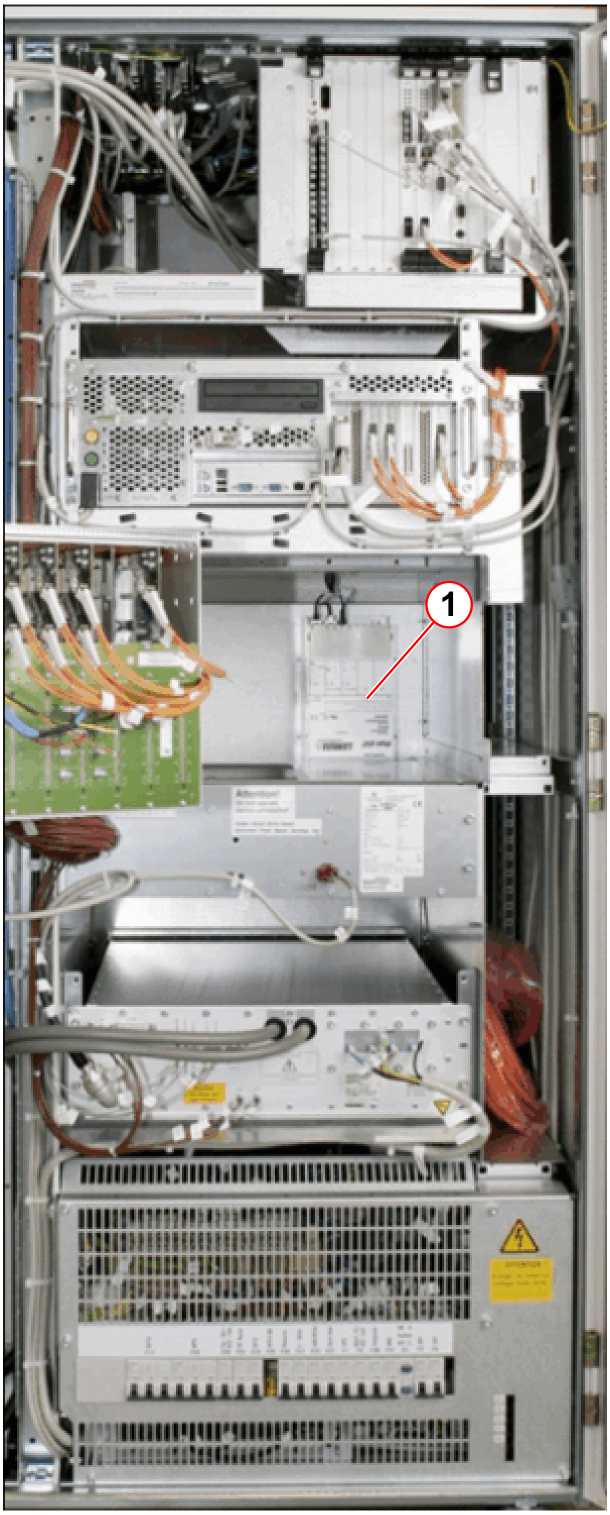
60 min

3.2.1.2 Tools

Normal tool set

3.2.2 Preliminary

Fig. 5: Line Power Distributor Parts Location



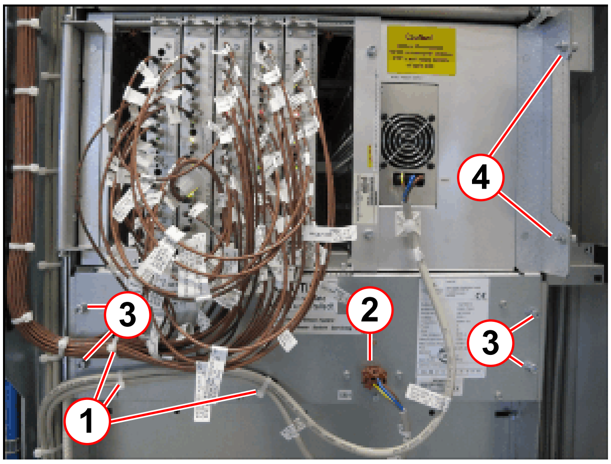
(1) Power supply (Lambda Vega 450)

The power supply is located behind the RFSU.

- Switch off the MR system
- Switch off F100 on the mains box

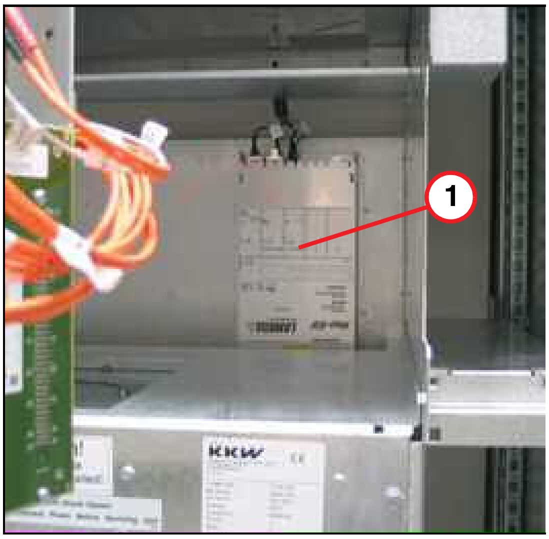
- Switch off ACC / F2

Fig. 6: RFSU and Air Distributor



- Remove the cable ties at the air distributor (Pos. 1)
- Remove the plug (Pos. 2)
- Remove the 4 screws at the air distributor (Pos. 3)
- Remove the RFSU screws (Pos. 4)

Fig. 7: Lambda VEGA 450



(1) Power Supply

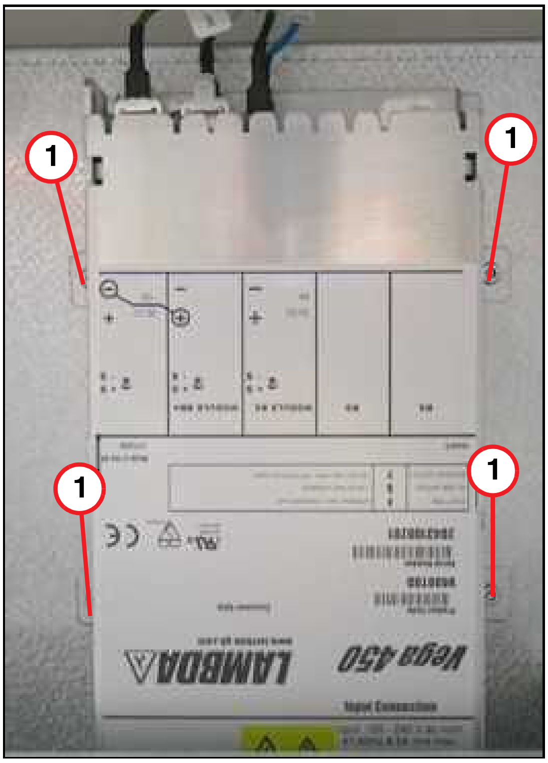
- Swivel out the RFSU frame

Fig. 8: Removed air distributor

- Push out the air distributor

3.2.3 Disassembly

Fig. 9: Lamda VEGA 450

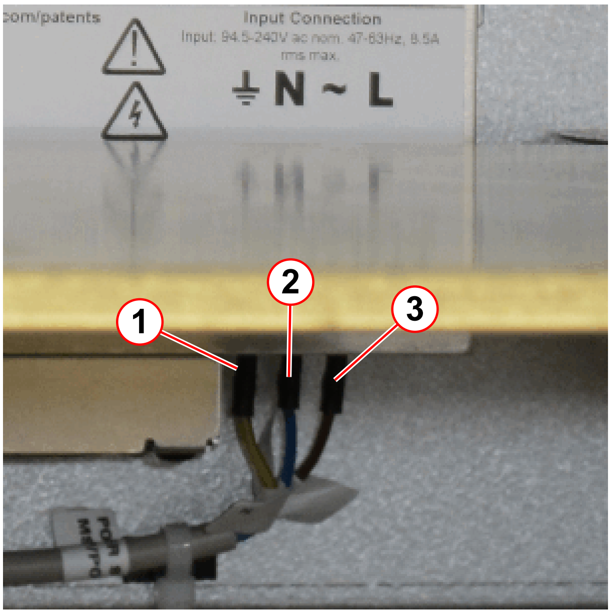


(1) Mounting screws

- Remove the cables
- Remove the mounting screws

3.2.4 Assembly

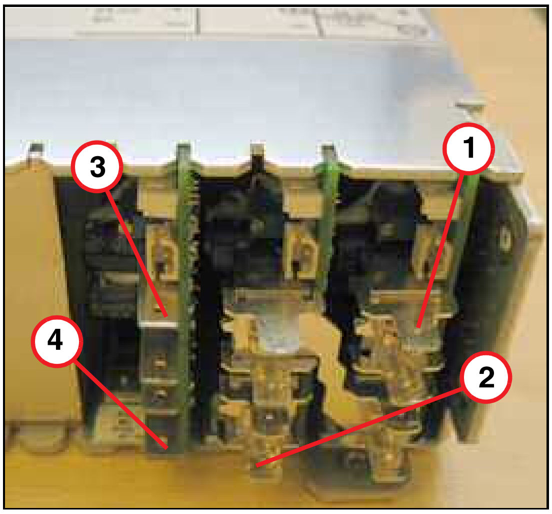
Fig. 10: PSU connection



(1) Ground (yellow)
(2) N (blue)
(3) L (brown)

- Install the new PSU
- Reconnect the cables

Fig. 11: PSU connections



(1) 36 V
(2) GND 36 V
(3) 24 V
(4) GND 24 V

3.2.5 Concluding steps

- Push in the air distributor unit.

- Swing in the swivel frame
- Tighten the screws on the RFSU and air distributor.
- Reconnect plug X801.
- Secure the cable with cable ties.
- Switch on ACC/F2.
- Switch on F100.

Version 02

- Chapter 3, Circuit breakers & contactors (➔ Fig. 4 Page 9) added.

Version 03

- Hint: "Protective earth connection test" was added.
- Chapter FRU Overview:
  Editorial changes.
- Chapter ACC:
  Editorial changes.

Version 04

- Chapter 3, Power Supply (Lambda Vega 450) reworked (MR_Charm_00455596).

There are no Hazard IDs in this document.

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Siemens Healthineers Headquarters
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Telephone: +49 9131 84-0
siemens.com/healthineers

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